

Permutation-Schedule Dominance for Two-Stage Assembly Flow Shops with Divisible Work

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Abstract

Permutation schedules are known to dominate in the classical two-stage assembly flow shop for makespan and weighted completion time. We extend this dominance property to a divisible-work model in which each stage-1 machine contains k identical parallel workers, job workloads may differ by machine, and work is allocated integrally across workers. The proof rests on a three-step logical chain. First, a capacity bound establishes that k workers can complete at most kt units of work by time t , limiting the total work completable before any deadline. Second, a water-filling allocation that balances worker loads achieves this bound with equality (up to an integer ceiling), so the constructed permutation schedule meets every release-time deadline set by the original schedule. Third, because release times do not increase, the max-plus recurrence at stage 2 yields pointwise domination: $C_j^{(2)}(S^*) \leq C_j^{(2)}(S)$ for every job j . This implies optimality of permutation schedules for every regular stage-2 objective, including makespan, total weighted completion time, and maximum lateness. The result extends to instances where each job visits only a subset of stage-1 machines. The classical model is recovered when $k = 1$. Small instances are exhaustively verified computationally, and the core arguments are formalized in Lean 4. We systematically map the boundaries of this dominance: it extends to two-stage chains but fails for chains of length $L \geq 3$, where we identify the precise failure mechanism. Computational evidence across over 20 million instances characterizes behavior for wider DAG topologies.

Keywords: assembly flow shop; permutation schedule; divisible work; weighted completion time; scheduling theory

Mathematics Subject Classification: 90B35

1 Introduction

The two-stage assembly flow shop, introduced by Lee et al. (1993), consists of m parallel machines at stage 1, each processing all n jobs without preemption, followed by a single assembly machine at stage 2 that can begin job j only after all stage-1 machines have completed j .

Potts et al. (1995) proved the problem is NP-hard for $m \geq 2$ and that *permutation schedules dominate*: the search for a $C_{\max}$ -optimal schedule may be restricted to schedules where all stage-1 machines use the same job ordering, reducing the search space from $(n!)^m$ orderings to $n!$. Tozkan et al. (2003) extended permutation-schedule dominance to the total weighted completion-time objective. Hariri and Potts (1997) developed a branch-and-bound algorithm with dominance theorems, testing on instances with up to $n = 8000$ jobs and $m = 10$ machines. Hadda et al. (2024) contributed new dominance rules and an exact branch-and-bound, solving instances up to $n = 1000$, $m = 8$.

Koulamas and Kyparisis (2007) studied a variant with *uniform* parallel machines at stage 1 (machines with different speeds). Their model differs from ours: uniform machines have one processor per machine at varying speeds, whereas our model has k identical workers per machine sharing divisible work.

We distinguish several adjacent lines of work. The malleable-job literature (Drozdowski, 2009) studies divisible work on parallel processors but without the assembly constraint. Al-Anzi and Allahverdi (2007) study the two-stage assembly problem with setup times, using permutation dominance to restrict the heuristic search space. Hatami et al. (2013) consider distributed assembly permutation flow shops (multiple factories, not multiple workers per machine). Zhang and Chen (2022) study assembly scheduling with parallel teams and heterogeneous worker skills, in a multi-objective setting without dominance results. Komaki et al. (2019) survey assembly flow shop models; the divisible-work model studied here does not appear in their taxonomy. Framinan et al. (2019) provide a comprehensive classification of concurrent-type assembly scheduling models, complementing the survey of Komaki et al. Fernandez-Viagas et al. (2022) develop speed-up procedures for insertion-based heuristics for the classical assembly flow shop, proving theoretical properties of the problem structure. These recent contributions address algorithmic and computational aspects of the classical model but do not establish dominance results for the divisible-work generalization.

To the best of our knowledge, the divisible-work assembly flow shop model — combining the assembly constraint with team-based parallel workers sharing divisible work — has not been studied previously, and the dominance result proved here is new.

In modern enterprise software engineering, cross-functional delivery frequently relies on parallel build phases (backend, frontend, platform teams) followed by a strict integration phase — precisely

the fork-join structure captured by the assembly flow shop. When teams independently optimise their own orderings, delivery stalls arise not from capacity shortages but from misaligned sequencing: one team completes its portion of an initiative while another has not yet started, leaving work blocked at the integration barrier.

While permutation-schedule dominance is highly desirable, a natural theoretical question is how far this property extends beyond the two-stage assembly topology. We conduct a systematic boundary investigation, proving that dominance extends to two-stage chains (Theorem 19) but fails for longer chains (Proposition 20), and providing extensive computational evidence for wider DAG structures. The complete dominance landscape is summarized in Table 1.

1.1 Contribution

We prove that permutation-schedule dominance extends to a divisible-work generalization where each stage-1 machine has $k \geq 1$ identical parallel workers and work requirements $w_{i,j}$ may differ across machines. Setting $k = 1$ recovers the classical model. The specific contributions are:

1. The extension from $k = 1$ to $k \geq 1$ workers per machine with integer work division, including the case $d_{i,j,\ell} \geq 0$ (some workers idle) and $w_{i,j} \geq 0$ (jobs may skip machines).
2. Machine-dependent work requirements $w_{i,j}$.
3. A constructive proof avoiding pairwise exchanges, unlike the exchange-argument technique of Potts et al. (1995). Their proof transforms a schedule by pairwise job exchanges, tracking the assembly barrier at each step; ours constructs the permutation schedule directly from a single inequality without intermediate exchanges. The construction requires $O(n \log n)$ time to sort jobs by release time and $O(nk)$ time per machine for water-filling (each job’s work is distributed in $O(k)$ time by batched floor-and-remainder allocation), giving $O(n \log n + mnk)$ overall.

The reduction from $(n!)^m$ to $n!$ has practical significance beyond computational tractability: when teams independently optimise their own orderings, the resulting schedule can be significantly sub-optimal. The magnitude of this “cost of independent prioritisation” is quantified computationally in Section 4.

2 Model

Definition 1 (Divisible-Work Two-Stage Assembly Flow Shop). *An instance consists of:*

- n jobs, indexed $j \in [n] = \{1, \dots, n\}$.
- m stage-1 machines, each with k identical workers ($k \geq 1$).
- One stage-2 assembly machine. Stage 2 can begin job j only after all stage-1 machines have completed j ; it then processes j for $q_j \in \mathbb{Z}_{>0}$ time units.

- Work requirements $w_{i,j} \in \mathbb{Z}_{\geq 0}$ for each machine $i \in [m]$ and job $j \in [n]$. When $w_{i,j} = 0$, job j does not require machine i ; machine i does not constrain r_j .

The integer work requirements $w_{i,j} \in \mathbb{Z}_{\geq 0}$ model atomic task units common in practice — person-days, story points, or discrete manufacturing batches — where partial units cannot be meaningfully assigned.

Definition 2 (Schedule). A schedule $S = (\sigma, \mathbf{d}, \tau)$ consists of:

- Permutations $\sigma_i \in \mathfrak{S}_n$ for each machine i : machine i processes jobs in order $\sigma_i(1), \dots, \sigma_i(n)$.
- Work divisions $d_{i,j,\ell} \in \mathbb{Z}_{\geq 0}$ for each machine i , job j , worker $\ell \in [k]$, satisfying $\sum_{\ell} d_{i,j,\ell} = w_{i,j}$.
- A permutation $\tau \in \mathfrak{S}_n$ giving the stage-2 processing order.

The work division $\mathbf{d}$ is determined before execution begins: it specifies, for each machine i and job j , how many units each worker ℓ will process. All processing is non-preemptive: each worker processes its assigned portion $d_{i,j,\ell}$ of a job as a single contiguous block before starting the next job in the permutation order. When $d_{i,j,\ell} = 0$, worker ℓ performs no work on job j and is available for the next job immediately.

Definition 3 (Completion Times). Worker ℓ on machine i processes jobs sequentially in order σ_i . The cumulative finish time after position s is $F_{i,\ell,s} = \sum_{t=1}^s d_{i,\sigma_i(t),\ell}$. The completion time of job j on machine i is the time at which all workers who contribute to j have finished:

$$C_{i,j} = \max_{\ell: d_{i,j,\ell} > 0} F_{i,\ell,\sigma_i^{-1}(j)}.$$

When $w_{i,j} = 0$ (so every $d_{i,j,\ell} = 0$), we set $C_{i,j} = 0$.

The release time of job j to stage 2 is $r_j = \max_{i \in [m]} C_{i,j}$.

Stage 2 processes jobs non-preemptively in order τ on a single machine. The stage-2 completion time of job $\tau(s)$ is given by the max-plus recurrence

$$C_{\tau(s)}^{(2)} = \max(C_{\tau(s-1)}^{(2)}, r_{\tau(s)}) + q_{\tau(s)},$$

with $C_{\tau(0)}^{(2)} := 0$. The makespan is $C_{\max}(S) = C_{\tau(n)}^{(2)}$.

Definition 4 (Permutation Schedule). A schedule is a permutation schedule if $\sigma_1 = \sigma_2 = \dots = \sigma_m$.

In the three-field notation of Graham et al. (1979), the problem is $\text{FH2}(P_{m \times k}, 1) \mid \text{assembly, div} \mid C_{\max}$, where “div” denotes divisible integer work.

3 Main Result

Theorem 5 (Pointwise Stage-2 Domination). For every feasible schedule S , there exists a permutation schedule S^* — whose permutation depends on the release times of S — such that $C_j^{(2)}(S^*) \leq C_j^{(2)}(S)$ for every job j .

Corollary 6. *Permutation schedules are optimal for every regular (non-decreasing) function of stage-2 completion times, including $C_{\max}$, $\sum_j v_j C_j^{(2)}$ for any $v_j > 0$, and $\max_j (C_j^{(2)} - \bar{d}_j)$ for any due dates $\bar{d}_j$.*

The proof of Theorem 5 uses two lemmas. The assembly constraint — stage 2 cannot begin job j until every stage-1 machine has finished — is captured by a single capacity inequality, from which the construction follows.

Lemma 7 (Capacity Bound). *Let S be any schedule with release times r_j . Let π be any permutation sorting jobs by non-decreasing release time: $r_{\pi(1)} \leq \dots \leq r_{\pi(n)}$ (ties broken arbitrarily). Then for every machine i and every $s \in [n]$:*

$$W_i(s) := \sum_{t=1}^s w_{i,\pi(t)} \leq k \cdot r_{\pi(s)}.$$

Proof. In schedule S , machine i completes job $\pi(t)$ by time $C_{i,\pi(t)} \leq r_{\pi(t)} \leq r_{\pi(s)}$ for each $t \leq s$. Hence by time $r_{\pi(s)}$, machine i has completed all jobs in $\{\pi(1), \dots, \pi(s)\}$, whose total work is $W_i(s)$. Since k workers each contribute at most $r_{\pi(s)}$ time units: $W_i(s) \leq k \cdot r_{\pi(s)}$. $\square$

Lemma 8 (Water-Filling Feasibility). *Given machine i processing jobs in order π , water-filling — assigning integer portions of each job's work so that cumulative worker loads remain as balanced as possible — produces a valid non-preemptive work division with $\sum_{\ell} d_{i,\pi(s),\ell} = w_{i,\pi(s)}$ and achieves*

$$C_{i,\pi(s)}^* \leq \lceil W_i(s)/k \rceil \quad \text{for each position } s.$$

Proof. We prove a load-balancing invariant, then derive the completion-time bound.

Load balancing. By induction on s . The invariant: after allocating work for position s , the cumulative load of each worker ℓ — namely $F_{i,\ell,s} = \sum_{t=1}^s d_{i,\pi(t),\ell}$ — lies in $\{\lfloor W_i(s)/k \rfloor, \lceil W_i(s)/k \rceil\}$. That is, loads differ by at most 1 and sum to $W_i(s)$.

Base case ($s = 0$): all loads are zero.

Inductive step: at position s , distribute $w_{i,\pi(s)}$ units one at a time, each to a currently least-loaded worker. Since the invariant holds at $s-1$, loads before allocation differ by at most 1. Adding units one-at-a-time to the minimum maintains the spread ≤ 1 : each unit either raises a minimum-load worker to the current maximum, or (once all workers are equal) raises all loads uniformly. The cumulative total becomes $W_i(s)$, so the maximum worker load is $\lceil W_i(s)/k \rceil$.

When $w_{i,\pi(s)} \geq k$, every worker receives at least one unit. When $w_{i,\pi(s)} < k$, some workers are idle for this job ($d_{i,\pi(s),\ell} = 0$).

Completion-time bound. By Definition 3, $C_{i,\pi(s)}^* = \max_{\ell: d_{i,\pi(s),\ell} > 0} F_{i,\ell,s}$. Every worker who contributes to job $\pi(s)$ has cumulative load $F_{i,\ell,s} \leq \lceil W_i(s)/k \rceil$ by the invariant. Hence $C_{i,\pi(s)}^* \leq \lceil W_i(s)/k \rceil$.

Non-preemptive feasibility. Worker ℓ processes jobs in the fixed order π : for each $s = 1, \dots, n$, the worker executes its $d_{i,\pi(s),\ell}$ units as a single contiguous block (taking zero time when $d_{i,\pi(s),\ell} = 0$). Since the job order is the same for all workers and each worker processes its portion without interruption, the schedule is non-preemptive in the sense of Definition 2. $\square$

Proof of Theorem 5. Given schedule S with release times r_j , let π sort jobs by non-decreasing r_j . Construct S^* :

1. Every stage-1 machine uses permutation π .
2. Each machine i independently uses water-filling (Lemma 8) to allocate $w_{i,\pi(s)}$.
3. Stage 2 uses the same order τ as S .

By Lemma 8, $C_{i,\pi(s)}^* \leq \lceil W_i(s)/k \rceil$. By Lemma 7, $W_i(s) \leq k \cdot r_{\pi(s)}$. Since $r_{\pi(s)}$ is a non-negative integer: $\lceil W_i(s)/k \rceil \leq r_{\pi(s)}$ for all i, s . Therefore $r_j^* \leq r_j$ for all j .

Since $r_j^* \leq r_j$ and stage 2 uses the same order τ , the max-plus recurrence gives, by induction on positions: $C_{\tau(s)}^{(2)}(S^*) \leq C_{\tau(s)}^{(2)}(S)$ for all s . Therefore $C_j^{(2)}(S^*) \leq C_j^{(2)}(S)$ for every j . The construction requires $O(n \log n)$ time to sort and $O(nk)$ per machine for water-filling—each job with work w assigns $\lfloor w/k \rfloor$ to every worker and distributes the $w \bmod k$ remainder one-per-worker among the minimum-load workers in $O(k)$ time, preserving the invariant that loads differ by at most 1—giving $O(n \log n + mnk)$ overall. $\square$

Remark 9 (Classical recovery). *When $k = 1$, water-filling is the identity and the theorem recovers the permutation-schedule dominance of Potts et al. (1995).*

Remark 10 (Existential nature and scope). *The result is a dominance statement, not a claim of global optimality over all permutations. The constructed permutation π depends on the release times of the given schedule S ; different schedules may yield different permutations. Theorem 5 asserts that for every feasible schedule, some permutation schedule is at least as good pointwise. It does not identify which permutation is globally optimal — that remains NP-hard.*

Remark 11 (Stage-2 monotonicity). *The final step uses a standard property of the max-plus recurrence: if $r_j^* \leq r_j$ for all j and the stage-2 order is held fixed, then $C_j^{(2)}(S^*) \leq C_j^{(2)}(S)$ for all j . This holds immediately by induction for the classical single-machine stage 2 defined above. The result extends to any stage-2 model whose completion times are monotone non-decreasing in release times, but does not extend to models with sequence-dependent setup times or blocking constraints where this monotonicity may fail.*

Remark 12 (Tie-breaking in π). *When multiple jobs share the same release time, any tie-breaking order suffices. The capacity bound (Lemma 7) holds for all orderings consistent with non-decreasing release times, since it depends only on the set of jobs completed by each time point, not on the ordering within a tie class.*

Remark 13 (Integrality). *The model assumes integer work requirements $w_{i,j} \in \mathbb{Z}_{\geq 0}$ and integer work divisions. The ceiling in the water-filling bound is needed when $w_{i,j} \geq 1$ on machines that process the job. In a continuous model (rational or real-valued work, arbitrary division), water-filling achieves exact balance $W_i(s)/k$ at each position, and the ceiling becomes an equality. The dominance result therefore holds a fortiori in the continuous case.*

Remark 14 (Job-dependent machine subsets). *Since $w_{i,j} \geq 0$ (Definition 1), the model accommodates jobs that visit only a subset $S_j = \{i : w_{i,j} > 0\} \subseteq [m]$ of stage-1 machines. Machine i processes only jobs in $J_i = \{j : w_{i,j} > 0\}$. The capacity bound (Lemma 7) holds on J_i : the cumulative work $W_i(s)$ sums only over jobs in J_i completed by time $r_{\pi(s)}$, and the k workers contribute at most $k \cdot r_{\pi(s)}$ units regardless of which jobs they process. Water-filling (Lemma 8) applies to the restricted subsequence. The proof is unchanged: $r_j^* = \max_{i \in S_j} C_{i,j}^* \leq r_j$.*

Remark 15 (Stage-2 re-optimization). *The construction inherits the stage-2 order τ unchanged. Re-optimizing τ for the improved release times can only improve the result but is not needed for dominance.*

Example. Let $n = 3$, $m = 2$, $k = 2$. Machine 1: $(w_{1,A}, w_{1,B}, w_{1,C}) = (6, 4, 2)$; machine 2: $(w_{2,A}, w_{2,B}, w_{2,C}) = (2, 6, 4)$.

Schedule S : machine 1 uses A, B, C ; machine 2 uses B, C, A . Water-filling gives: machine 1 completes A at 3, B at 5, C at 6; machine 2 completes B at 3, C at 5, A at 6. Release times: $r_A = 6$, $r_B = 5$, $r_C = 6$.

Sort: $\pi = (B, A, C)$. Capacity bound holds: $W_1(1) = 4 \leq 10$, $W_2(1) = 6 \leq 10$, etc. Water-filling with π : $r_B^* = 3 \leq 5$, $r_A^* = 5 \leq 6$, $r_C^* = 6 \leq 6$. Figure 1 shows the two schedules side by side.

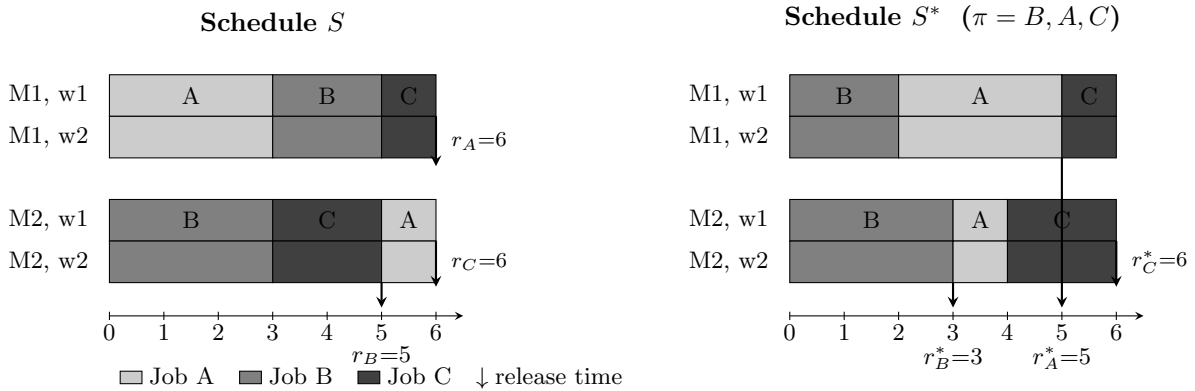


Figure 1: Worked example: unsynchronized schedule S (left) vs. synchronized schedule S^* (right). Each machine has $k = 2$ workers shown as two rows. Release times improve or stay equal for every job: $r_B^* = 3 \leq 5 = r_B$, $r_A^* = 5 \leq 6 = r_A$, $r_C^* = 6 \leq 6 = r_C$.

4 Computational Verification and Lean 4 Formalization

We verified Theorem 5 by exhaustive enumeration across 27 homogeneous-work configurations ($w_{i,j} = w_j$; 3,846,136 schedules) and 12 heterogeneous-work configurations ($w_{i,j}$ varying across machines; 167,120 schedules), with $n \in \{2, 3, 4, 5\}$, $m \in \{2, 3\}$, $k \in \{2, 3\}$. The homogeneous configurations include symmetric work (w_j equal for all j), asymmetric work (w_j varying across jobs), and value-weighted variants (with non-uniform $v_j > 0$) to test the weighted completion-time objective. For every schedule S in every configuration, the construction of Theorem 5 produced a permutation schedule S^* satisfying $C_j^{(2)}(S^*) \leq C_j^{(2)}(S)$ for every job j — i.e., pointwise stage-2 domination was verified exhaustively.

Beyond verifying dominance, we quantified the practical cost of independent prioritisation. Across thousands of random configurations with $n \in \{3, \dots, 8\}$, $m \in \{2, 3, 4\}$, $k \in \{2, 3\}$, and work values drawn uniformly from $\{1, \dots, 6\}$, a randomly chosen permutation schedule achieves equal or better makespan than a randomly chosen independent schedule in 96–99.9% of instances, with an average makespan improvement of 17–29%. This gap widens with n , reflecting the combinatorial explosion of misalignment opportunities as the number of jobs grows. These figures compare two equally imperfect selection processes; locally optimised independent orderings would narrow the gap, but cannot eliminate it by Theorem 5.

The dominance argument and the water-filling feasibility argument (Theorem 5) are formalized in Lean 4 (v4.29.0, Mathlib) with zero unresolved axioms.¹ The $C_{\max}$ corollary’s bubble-sort composition lemma and supplementary interchange-argument lemmas (an alternative proof path) carry `sorry` axioms; these are not used by Theorem 5.

5 Extension to Multi-Stage Chains

Having established dominance for the assembly topology in Sections 3–4, we now ask: how far does this property extend? The two-stage assembly model assumes independent parallel machines at stage 1. In many applications, work flows through a *chain* of dependent stages: stage ℓ cannot begin job j until stage $\ell-1$ completes j .

To analyse multi-stage chains and intermediate starvation mechanisms, we transition to a continuous-time, preemptive priority model. As noted in Remark 13, the two-stage dominance result holds *a fortiori* in the continuous case, so this modelling choice does not weaken the main result. The preemptive discipline captures settings where teams respond immediately to arriving high-priority work, the natural operating mode in cross-functional delivery.

We develop the answer in four steps. First, a single-stage makespan invariance lemma (Lemma 18) shows that the local priority ordering does not affect the makespan at any one stage, which immediately yields a positive result for $L = 2$ chains (Theorem 19). Second, for $L \geq 3$ we exhibit a minimal counterexample showing that pointwise domination fails: unsynchronised schedules can exploit preemption at intermediate stages to beat every synchronised alternative. Third, we strengthen the

¹Source code, verification data, and Lean formalization: <https://github.com/ebowman/prioritizer>.

negative result to makespan and total weighted completion time (Proposition 20). Fourth, we identify a restricted class of *concordant* work profiles under which synchronised schedules recover flowtime dominance for $n = 2$ jobs (Theorem 23).

Definition 16 (*L-Stage Chain with Preemptive Priority*). *An L-stage chain consists of stages $1, 2, \dots, L$, each with $k_\ell \geq 1$ identical workers and work requirements $w_{\ell,j} \in \mathbb{Z}_{>0}$. Jobs are released to stage 1 at time 0. Stage ℓ ($\ell \geq 2$) can begin job j only after stage $\ell-1$ completes j .*

A preemptive priority schedule is defined by a single permutation π (the priority ordering). At each stage, when a worker becomes free, it begins (or resumes) the highest-priority job that has arrived and is not yet complete at that stage. A job in progress may be preempted when a higher-priority job arrives from the preceding stage; the preempted job is resumed when no higher-priority work remains. Within each job, work is divided across the k_ℓ workers by water-filling.

Time model. Unlike the integer-division model of Sections 2–3, processing times in the chain model are $w_{\ell,j}/k_\ell$, which may be non-integer even when work requirements $w_{\ell,j}$ are integral. All completion-time calculations in this section use continuous (real-valued) time. Since the two-stage dominance result holds *a fortiori* in the continuous case (Remark 13), this does not weaken the two-stage result.

Under preemptive priority, higher-priority jobs flow through the chain without waiting: the highest-priority job is never preempted and completes each stage as soon as capacity allows. This gives a key structural property.

Lemma 17 (Priority-Order Completion). *Under a preemptive priority schedule with ordering π , jobs complete each stage in priority order: $C_\ell^P(\pi(1)) \leq C_\ell^P(\pi(2)) \leq \dots \leq C_\ell^P(\pi(n))$ for every stage ℓ .*

Proof. By induction on ℓ . At stage 1, all jobs are released at time 0. Preemptive priority ensures $\pi(1)$ is never preempted and completes first; $\pi(2)$ is preempted only by $\pi(1)$ and completes second; and so on.

At stage $\ell \geq 2$, jobs arrive in priority order by the inductive hypothesis (since $C_{\ell-1}^P$ is in priority order). The same preemptive argument applies: $\pi(i)$ is preempted only by $\pi(1), \dots, \pi(i-1)$, all of which arrive earlier and complete earlier. Hence $\pi(i)$ completes in position i . $\square$

A key structural property of the preemptive priority model is that the makespan at any single stage is invariant to the priority ordering at that stage, depending only on the arrival times and work profile.

Lemma 18 (Single-Stage Makespan Invariance). *At a single stage with k workers under preemptive priority, the makespan (completion of the last job) depends only on the job arrival times and work requirements, not on the priority ordering.*

Proof sketch. The key observation is that preemptive priority scheduling is *work-conserving*: all k workers are busy whenever any job is available, so the stage has a binary state—active (processing

at rate k) or idle (no work present). Since the cumulative work available at any time depends only on arrival times, not priority ordering, the active/idle structure is invariant across permutations, and so is the makespan. The full argument is given in Appendix A. $\square$

Theorem 19 (Two-Stage Chain Makespan Dominance). *For $L = 2$, the optimal makespan over all per-stage orderings equals the optimal makespan over synchronised (single-permutation) schedules.*

Proof. Consider any schedule with orderings π_1, π_2 at stages 1 and 2 respectively. By Lemma 18, the makespan at stage 2 depends only on the arrival times at stage 2 (the completion times from stage 1) and the work profile at stage 2, not on π_2 . Therefore the synchronised schedule (π_1, π_1) achieves the same makespan as (π_1, π_2) . Since this holds for every π_1 , the minimum over synchronised schedules equals the minimum over all schedules. $\square$

For longer chains ($L \geq 3$), the natural inductive extension—sort by C_{L-1}^S , apply the $(L-1)$ -stage result, then use stage- L monotonicity—requires that the ordering good for stages $1, \dots, L-1$ is also good for stage L . **This is not always the case.** When work requirements are strongly asymmetric across stages (e.g., job j is heavy at stage ℓ but light at stage $\ell+1$), different stages benefit from different orderings, and no single permutation dominates.

Exhaustive computation confirms that $L \geq 3$ chains with inverted work profiles can admit unsynchronised schedules that are Pareto-superior to every synchronised one *pointwise*. A minimal counterexample for pointwise dominance has $L = 3, n = 2, k = 2$, with work $w = [(5, 2), (2, 5), (4, 3)]$: the unsynchronised ordering $[1, 0] \mid [0, 1] \mid [0, 1]$ achieves $(6.5, 7.0)$ at stage 3, while the synchronised orderings achieve $(5.5, 7.5)$ and $(7.0, 5.0)$ — neither dominates pointwise.

Pointwise domination failing implies that dominance fails for *some* regular objectives. In the example above, total tardiness with due dates $d = (6.5, 7.0)$ gives $T_d = 0$ for the unsynchronised schedule but $T_d = 0.5$ for both synchronised orderings.

More significantly, dominance also fails for makespan and total weighted completion time at $L \geq 3$.

Proposition 20 (Failure of chain dominance for $L \geq 3$). *For $L \geq 3$, there exist instances where no synchronised schedule achieves the optimal makespan. Similarly, there exist instances where no synchronised schedule achieves the optimal total weighted completion time.*

Proof. Counterexample: $L = 3, n = 2, k = 2, w = [(1, 2), (4, 1), (1, 2)]$. The processing times (work divided by k , in the continuous-time model) are $(0.5, 1.0), (2.0, 0.5), (0.5, 1.0)$.

Under the synchronised ordering $[0, 1]$: at stage 1, job 0 completes at $t = 0.5$; job 1 completes at $t = 1.5$. At stage 2, job 0 processes from $t = 0.5$ to 2.5 ; job 1 then processes from $\max(2.5, 1.5) = 2.5$ to 3.0 . At stage 3, job 0 arrives at 2.5 and completes at 3.0 ; job 1 arrives at 3.0 and completes at 4.0 . Makespan: 4.0 . Total flowtime: 7.0 .

Under the synchronised ordering $[1, 0]$: by symmetry of the argument, the makespan is also 4.0 and total flowtime 6.5 .

Under the unsynchronised ordering $[0, 1] \mid [1, 0] \mid \text{any}$ (the stage-3 ordering is irrelevant by Lemma 18, since stage 3 is the final stage): stage 1 (priority $[0, 1]$): both workers process job 0 first (completing at $t = 0.5$), then job 1 (completing at $t = 1.5$). At stage 2 (priority $[1, 0]$), job 0 arrives first at $t = 0.5$ and begins processing. Job 1 arrives at $t = 1.5$ with higher priority under $[1, 0]$ and preempts job 0 (which has completed 2 of 4 work units). Job 1 processes in 0.5 time, completing at $t = 2.0$. Job 0 resumes and completes at $t = 3.0$. At stage 3, job 1 arrives at 2.0 and completes at 3.0; job 0 arrives at 3.0 and completes at 3.5. Makespan: 3.5. Total flowtime: 6.5.

Hence the unsynchronised schedule achieves strictly better makespan ($3.5 < 4.0$) and equal-or-better flowtime. Exhaustive enumeration over all $2^3 = 8$ orderings confirms no synchronised schedule achieves makespan below 4.0.

For weighted completion time, the stage-3 completion vectors are $C^{[0,1]} = (3.0, 4.0)$, $C^{[1,0]} = (4.0, 2.5)$, and $C^{\text{unsync}} = (3.5, 3.0)$. With weights $v = (3, 2)$, the synchronised schedules give $\sum v_j C_j^{[0,1]} = 3(3.0) + 2(4.0) = 17.0$ and $\sum v_j C_j^{[1,0]} = 3(4.0) + 2(2.5) = 17.0$, while the unsynchronised schedule gives $\sum v_j C_j^{\text{unsync}} = 3(3.5) + 2(3.0) = 16.5 < 17.0$. Hence no synchronised schedule is optimal for $\sum v_j C_j$. $\square$

Remark 21 (Mechanism). *The counterexample exploits preemption at an intermediate stage. Job 0 is light at stage 1 (work 1) but heavy at stage 2 (work 4), while job 1 is heavy at stage 1 (work 2) but light at stage 2 (work 1). The unsynchronised schedule uses opposite priorities at stages 1 and 2: stage 1 prioritises job 0 (light), completing it quickly so it flows to stage 2 early. At stage 2, job 1 has higher priority and preempts job 0 when it arrives, completing its short stage-2 work immediately. This pairs the heavier stage-2 job (job 0) with an earlier start, reducing idle time. A synchronised schedule cannot achieve this because it must use the same priority at all stages.*

Systematic search over 46,656 configurations ($L = 3$, $n = 2$, $k \in \{2, 3\}$, work values 1–6) found 1,414 makespan counterexamples and 1,016 flowtime counterexamples. The largest makespan gap is 1.0 (e.g., $\text{sync} = 5.5$, $\text{unsync} = 4.5$).

Remark 22 (Scope of the failure). *The failure requires $L \geq 3$ for makespan. By Theorem 19, synchronised schedules are optimal for makespan at $L = 2$. The failure requires work profiles where the relative difficulty of jobs changes across stages (“inverted” profiles), enabling preemption at intermediate stages to reshuffle the completion order.*

Under a restricted class of work profiles, synchronised schedules *do* dominate for total flowtime. Call work profiles *concordant* if, for every pair of jobs i, j , the ordering $w_{\ell,i}$ versus $w_{\ell,j}$ is the same at every stage where they differ.

Theorem 23 (Concordant Flowtime Dominance, $n = 2$). *For any L -stage chain with preemptive priority, two concordant jobs, and $k_\ell \geq 1$ workers per stage, the synchronised schedule that places the lighter job first at every stage minimises total flowtime $\sum_j C_j^L$.*

Proof sketch. Let $p_\ell \leq q_\ell$ be the processing times of jobs A (lighter) and B (heavier) at stage ℓ . Define the *sync value* $\Phi_L(\alpha, \beta)$ for arrivals $\alpha \leq \beta$ by the recursion $\Phi_L(\alpha, \beta) = \Phi_{L-1}(\alpha + p_1, \max(\alpha + p_1, \beta) + q_1)$ with $\Phi_0(\alpha, \beta) = \alpha + \beta$.

Two properties of Φ : (i) *componentwise monotonicity*: Φ_L is non-decreasing in each argument; (ii) *sum-preserving monotonicity*: at fixed sum $T = \alpha + \beta$, $\Phi_L(\alpha, T - \alpha)$ is non-decreasing in α .

The proof uses two interlocking inductions on L . *Proposition (P)*: for $\alpha \leq \beta$, any schedule gives flowtime $\geq \Phi_L(\alpha, \beta)$. *Lemma (Q)*: for $a > b$, any schedule gives flowtime $\geq \Phi_L(b, a)$.

At each stage, three cases arise (no overlap, overlap with A-first, overlap with B-first). In each, the stage-1 outputs feed the $(L-1)$ -stage inductive hypothesis for (P) or (Q), and the comparison reduces to monotonicity or sum-preserving monotonicity of Φ . The full proof, including all three cases at each stage and both inductive propositions, appears in Appendix B. The result has been verified computationally for all concordant $n = 2$ configurations with $L \leq 3$, $k = 2$, work values 1–8. $\square$

The theorem is verified exhaustively for $n = 2$ with $L \leq 3$, $k = 2$, and work values 1–8 (21,952 strictly concordant configurations, 0 flowtime failures).

The result does *not* extend to $n \geq 3$, even under strict concordance. A counterexample: $L = 2$, $n = 3$, $k = 2$, $w = [(4, 5, 6), (1, 9, 10)]$. The synchronised lighter-first schedule gives flowtime 25.5; the unsynchronised $[1, 0, 2] \mid [0, 1, 2]$ achieves 25.0 by preempting a medium-weight job with the lightest one at stage 2. Among 14,400 strictly concordant $n = 3$, $L = 2$ configurations with work values 1–10, this is the sole counterexample, indicating that failures require large work asymmetry. Exhaustive search over 175,616 strictly concordant $n = 3$, $L = 3$ configurations with work values 1–8 found 0 flowtime failures.

Remark 24 (Operational interpretation). *Sections 3–4 show that the cost of independent prioritisation can be completely eliminated for two-stage assembly: sharing a single ranked list of initiatives is optimal for every regular objective. Theorem 19 extends this to two-stage chains. For n initiatives across m teams, the search space contracts from $(n!)^m$ independent orderings to $n!$ shared orderings, and computational experiments confirm that the practical benefit of alignment is robust to imperfect ranking.*

For deeper chains ($L \geq 3$), independent prioritisation is no longer universally harmful. When teams have strongly differing workloads per initiative, an intermediate team may benefit from processing a short arriving task ahead of a long one, even if the short task has lower global priority, because doing so delivers work to the downstream team sooner. However, exploiting this benefit requires preemption at intermediate stages — operationally, the ability to interrupt work in progress when higher-priority items arrive — which is itself costly and complex.

6 Conclusion

For any schedule in the divisible-work two-stage assembly flow shop, we have constructed a permutation schedule with $C_j^{(2)}(S^*) \leq C_j^{(2)}(S)$ for every job j , using a capacity bound and water-filling (Theorem 5). This extends the classical results of Potts et al. and Tozkapan et al. from $k = 1$ to arbitrary k , with machine-dependent work requirements.

For L -stage chains with preemptive priority, a single-stage makespan invariance lemma (Lemma 18) shows that the makespan at any stage depends only on arrival times and work, not on the local priority ordering. This yields a short proof that synchronised schedules are makespan-optimal for $L = 2$ chains (Theorem 19). However, for $L \geq 3$, the invariance applies only to the final stage; at intermediate stages, the priority ordering affects *individual* completion times that serve as arrival times downstream. A minimal counterexample (Proposition 20: $L = 3$, $n = 2$, $k = 2$, $w = [(1, 2), (4, 1), (1, 2)]$) shows that unsynchronised schedules can exploit preemption at intermediate stages to achieve strictly better makespan (3.5 vs. 4.0) and flowtime.

Table 1 summarizes the dominance results across model variants, making the sharp boundaries visible.

Table 1: Dominance landscape across model variants.

Model	Property	Result
Two-stage assembly (Thm. 5)	Pointwise domination (all regular obj.)	✓ Holds
Job-dependent subsets (Rem. 14)	Pointwise domination	✓ Holds
Two-stage chain, $L = 2$ (Thm. 19)	Makespan optimality	✓ Holds
L -stage chain, $L \geq 3$ (Prop. 20)	Pointwise domination	— Fails
L -stage chain, $L \geq 3$ (Prop. 20)	Makespan optimality	— Fails
Concordant $n = 2$ chain (Thm. 23)	Flowtime optimality	✓ Holds
<i>Computational evidence (not proven):</i>		
Diamond ($S \rightarrow \{A, B\} \rightarrow T$), $n = 2$	Pointwise domination	✓* Holds
Diamond, $n \geq 3$	Makespan optimality	✓* Holds
FJFJ ($\{A, B\} \rightarrow C \rightarrow \{D, E\} \rightarrow F$)	Makespan optimality	✓* Holds
Tree ($A \rightarrow \{B, C\}$, $B \rightarrow \{D, E\}$), $n \geq 3$	Makespan optimality	—* Fails (rare)

*Computational evidence: tested across 10^7 + random instances with zero failures (diamond/FJFJ) or rare failures (tree: $\sim 0.02\%$). Not a proof.

The core capacity bound and water-filling feasibility arguments for Theorem 5 are verified in Lean 4 with zero unresolved axioms, providing a higher level of assurance than computational verification alone.

To state the computational results precisely, we define the *DAG scheduling model*. A directed acyclic graph $G = (V, E)$ specifies the dependency structure among stages. Each node $v \in V$ has k_v identical parallel workers, and each job j requires $w_{v,j}$ units of work at node v . Job j is released to node v only after all predecessor nodes have completed j . Within each stage, the preemptive priority discipline from Section 5 is used: all k_v workers concentrate on the highest-priority available job. A *permutation schedule* (called “synchronised” above) uses the same job ordering π at every node.

We examine three specific topologies:

- **Diamond.** $V = \{S, A, B, T\}$, $E = \{S \rightarrow A, S \rightarrow B, A \rightarrow T, B \rightarrow T\}$. The source S and sink T are assembly points; A and B execute in parallel. This models a foundation or architecture phase feeding parallel build teams, followed by integration.

- **Fork-join-fork-join (FJFJ).** $V = \{A, B, C, D, E, F\}$, $E = \{A \rightarrow C, B \rightarrow C, C \rightarrow D, C \rightarrow E, D \rightarrow F, E \rightarrow F\}$. Two fork-join stages in sequence; sources A, B feed join node C , which forks to D, E , reconverging at sink F . This captures alternating rounds of parallel build and integration checkpoints.
- **Tree.** $V = \{A, B, C, D, E\}$, $E = \{A \rightarrow B, A \rightarrow C, B \rightarrow D, B \rightarrow E\}$. Asymmetric branching without reconvergence; sinks are C, D , and E . This represents asymmetric dependency structures where downstream teams do not reconverge.

Note that all DAG results use the *preemptive* priority model introduced in Section 5, not the non-preemptive model of Section 2.

We have explored computationally whether dominance extends beyond assembly and chain topologies to these DAG dependency structures. For *diamond* DAGs, pointwise domination holds for $n = 2$ but fails (rarely) for $n \geq 3$; however, *makespan* dominance holds in all 262,144 tested instances for $n = 2$ (exhaustive over work values 1–8 at the three non-source stages, $k = 2$) and over 10 million random instances for $n \geq 3$ ($n \in \{3, 4, 5\}$, work values 1–6, $k \in \{2, 3\}$), with zero failures. For *FJFJ* DAGs, pointwise domination fails for $n \geq 2$, but makespan dominance holds across all 9,765,625 tested instances with zero failures (exhaustive over work values 1–5 at the five non-source stages, $n = 2$, $k = 2$). The only structure where makespan dominance fails is *tree* DAGs, where rare counterexamples appear for $n \geq 3$ (approximately 1 in 5,000 random instances; $n \in \{3, 4, 5\}$, work values 1–6, $k \in \{2, 3\}$). The common pattern is that parallel width at intermediate stages—present in diamond and FJFJ but absent in trees—appears to rescue makespan dominance even when pointwise domination fails. We conjecture that permutation schedules are makespan-optimal for diamond and FJFJ topologies, supported by over 20 million computationally verified instances across these structures, though formal proofs remain open. Stochastic processing times also remain open. The integer model for $L \geq 3$ chains may also merit investigation: the counterexamples in Section 5 use continuous-time processing, and it is unknown whether integrality constraints could rescue dominance for longer chains.

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gation, Writing – original draft, Writing – review & editing, Visualization.

A Proof of Lemma (Single-Stage Makespan Invariance)

Proof. Consider a single stage with k identical workers operating under preemptive priority. Jobs arrive at times $a_1 \leq a_2 \leq \dots \leq a_n$ with work requirements $w_1, w_2, \dots, w_n$ (each divisible into k equal parts). Let π be any priority permutation of the jobs.

Work-conserving property. Under preemptive priority, all k workers are assigned to the highest-priority available job. In particular, every worker is busy whenever at least one job is present in the stage. The stage therefore has exactly two states: *active* (processing at aggregate rate k) and *idle* (no work available). This binary behaviour holds regardless of the chosen permutation π .

Cumulative work available. Define $W(t) = \sum_{j: a_j \leq t} w_j$ as the cumulative work that has arrived by time t . This quantity depends only on the arrival times and work sizes, not on π .

Active/idle structure is π -invariant. Let $A(t)$ denote the cumulative active time up to time t . The work consumed by time t is $k \cdot A(t)$. Because the stage is work-conserving, the stage is idle at time t if and only if all arrived work has been consumed: $k \cdot A(t) = W(t)$, and is active otherwise. Since $W(t)$ is independent of π , and $A(t)$ is determined by the condition $k \cdot A(t) = \min(W(t), k \cdot t')$ where t' is the elapsed non-idle time, the function $A(t)$ is itself independent of π . Consequently, the set of idle intervals is identical for every permutation.

Makespan invariance. Let $W = \sum_j w_j$ be the total work and let I be the total idle time. Then

$$C_{\max} = \frac{W}{k} + I.$$

Since both W/k and I are independent of π , the makespan $C_{\max}$ is invariant across all priority orderings. $\square$

B Proof of Theorem 23: Concordant Flowtime Dominance

This appendix gives the complete proof of Theorem 23. We restate the setting and the theorem for convenience, then develop the argument in three parts: single-stage analysis, the sync value function and its monotonicity properties, and the main inductive proof.

B.1 Model and statement

An L -stage chain has stages $1, \dots, L$, each with $k_\ell \geq 1$ identical workers. Two jobs A (lighter) and B (heavier) have work requirements $w_\ell(A), w_\ell(B)$ at stage ℓ , with processing times $p_\ell = w_\ell(A)/k_\ell$ and $q_\ell = w_\ell(B)/k_\ell$. Under preemptive priority, all k_ℓ workers focus on the highest-priority available job; a job in progress is preempted when a higher-priority job arrives.

Work profiles are *concordant*: $p_\ell \leq q_\ell$ for every stage ℓ (job A is always lighter).

A *per-stage preemptive priority schedule* is a vector $\sigma = (\sigma_1, \dots, \sigma_L) \in \{A, B\}^L$ specifying which job has priority at each stage. The *synchronised* schedule is $\sigma^* = (A, A, \dots, A)$.

Jobs are released to stage 1 at time 0. Stage $\ell \geq 2$ can begin job j only after stage $\ell-1$ completes j .

Theorem 23 (restated). *For any L -stage concordant chain with two jobs, the synchronised schedule σ^* minimises total flowtime:*

$$C_L^{\sigma^*}(A) + C_L^{\sigma^*}(B) \leq C_L^\sigma(A) + C_L^\sigma(B)$$

for every per-stage preemptive priority schedule σ .

B.2 Single-stage analysis

Lemma A.1 (Single-stage transitions). At a single stage with $p \leq q$:

Case 1: $a_A \leq a_B$ (A arrives first).

(1a) *No overlap* ($a_A + p \leq a_B$): both priorities give $C(A) = a_A + p$, $C(B) = a_B + q$.

(1b) *Overlap, A-first* ($a_A + p > a_B$): $C(A) = a_A + p$, $C(B) = a_A + p + q$.

(1c) *Overlap, B-first* ($a_A + p > a_B$): $C(B) = a_B + q$, $C(A) = a_A + p + q$.

Case 2: $a_B < a_A$ (B arrives first).

(2a) *No overlap* ($a_B + q \leq a_A$): both priorities give $C(B) = a_B + q$, $C(A) = a_A + p$.

(2b) *Overlap, A-first* ($a_B + q > a_A$): $C(A) = a_A + p$, $C(B) = a_B + p + q$.

(2c) *Overlap, B-first* ($a_B + q > a_A$): $C(B) = a_B + q$, $C(A) = a_B + p + q$.

Proof. In each overlap case, the server is busy continuously from the first arrival to the last completion, processing total work $p + q$. Between the two arrivals, the first-arriving job is the only one available. After both have arrived, the priority determines which finishes first. $\square$

Lemma A.2 (Single-stage flowtime optimality). At a single stage with $p \leq q$ and $a_A \leq a_B$:

(i) A-first gives $C(A) + C(B) \leq C'(A) + C'(B)$ where primed denotes B-first.

(ii) Under A-first: $C(A) \leq C(B)$.

Proof. No overlap: both priorities give the same result, and $C(A) = a_A + p \leq a_B + q = C(B)$.

Overlap: A-first sum = $2a_A + 2p + q$; B-first sum = $a_A + a_B + p + 2q$. Difference $(a_A - a_B) + (p - q) \leq 0$. Under A-first, $C(A) = a_A + p < a_A + p + q = C(B)$. $\square$

B.3 Sync value function

Under the synchronised schedule σ^* with initial arrivals (α, β) , $\alpha \leq \beta$, job A is never preempted. By Lemma A.2(ii), A completes each stage before B , giving $C_\ell^*(A) = \alpha + \sum_{s=1}^\ell p_s$ and the recurrence $C_\ell^*(B) = \max(C_\ell^*(A), C_{\ell-1}^*(B)) + q_\ell$ with $C_0^*(B) = \beta$.

Definition A.3 (Sync value). For an L -stage concordant chain and initial arrivals (α, β) with $\alpha \leq \beta$, define $\Phi_L(\alpha, \beta) = C_L^*(A) + C_L^*(B)$. The recursion is

$$\Phi_L(\alpha, \beta) = \Phi_{L-1}(\alpha + p_1, \max(\alpha + p_1, \beta) + q_1)$$

with $\Phi_0(\alpha, \beta) = \alpha + \beta$.

Lemma A.4 (Componentwise monotonicity). $\Phi_L(\alpha, \beta)$ is non-decreasing in both α and β (for $\alpha \leq \beta$).

Proof. Induction on L . For $L = 0$: $\Phi_0 = \alpha + \beta$, non-decreasing in both. For $L \geq 1$: $\Phi_L(\alpha, \beta) = \Phi_{L-1}(f(\alpha, \beta), g(\alpha, \beta))$ where $f = \alpha + p_1$ and $g = \max(\alpha + p_1, \beta) + q_1$ are both non-decreasing in α and β . By the inductive hypothesis, the composition is non-decreasing. $\square$

Lemma A.5 (Sum-preserving monotonicity). For fixed sum $T = \alpha + \beta$ with $\alpha \leq \beta$: $\Phi_L(\alpha, T - \alpha)$ is non-decreasing in α .

Proof. Induction on L . For $L = 0$: $\Phi_0 = T$, constant. For $L \geq 1$:

$$\Phi_L(\alpha, T - \alpha) = \Phi_{L-1}(\alpha + p_1, \max(\alpha + p_1, T - \alpha) + q_1).$$

Both arguments are non-decreasing in α : the first $f(\alpha) = \alpha + p_1$ is strictly increasing; the second $g(\alpha) = \max(\alpha + p_1, T - \alpha) + q_1$ is piecewise linear, equalling $T - \alpha + q_1$ (decreasing) when $2\alpha + p_1 \leq T$ and $\alpha + p_1 + q_1$ (increasing) otherwise, so g is non-decreasing for $\alpha \geq (T - p_1)/2$.

For the decreasing regime of g : $f(\alpha) + g(\alpha) = \alpha + p_1 + T - \alpha + q_1 = T + p_1 + q_1$ (constant), and f is increasing, so by the inductive hypothesis applied to the sum $f + g = T + p_1 + q_1$ with first argument $f(\alpha)$ increasing, the composition is non-decreasing.

For the increasing regime of g : both f and g are non-decreasing, so by Lemma A.4 the composition is non-decreasing. $\square$

B.4 Main proof: simultaneous induction

We prove two statements simultaneously by induction on L .

Proposition (P) (A-first arrivals). For any L -stage concordant chain, arrivals (α, β) with $\alpha \leq \beta$, and any schedule σ : $\Phi_L(\alpha, \beta) \leq C_L^\sigma(A) + C_L^\sigma(B)$.

Lemma (Q) (B-first arrivals). For any L -stage concordant chain, arrivals (a, b) with $a > b \geq 0$, and any schedule σ : $C_L^\sigma(A) + C_L^\sigma(B) \geq \Phi_L(b, a)$.

Theorem 23 follows from Proposition (P) with $\alpha = \beta = 0$.

Lemma (Q) says: when B arrives before A , the flowtime is at least as large as the sync value from the sorted arrivals (b, a) . Intuitively, the “wrong” arrival order cannot help.

Base case: $L = 0$

$\Phi_0(\alpha, \beta) = \alpha + \beta = C_0(A) + C_0(B)$. Both statements hold with equality.

Base case: $L = 1$

Proposition (P): $\alpha \leq \beta, p_1 \leq q_1$. By Lemma A.2, A-first minimises flowtime.

Lemma (Q): $a > b$. We check each case from Lemma A.1.

Case 2a ($b + q_1 \leq a$): $\text{sum} = a + b + p_1 + q_1$. Since $b + p_1 \leq b + q_1 \leq a$: $\Phi_1(b, a) = (b + p_1) + (a + q_1) = a + b + p_1 + q_1$. Equality. ✓

Case 2b ($b + q_1 > a$, A-first): $\text{sum} = a + b + 2p_1 + q_1$. Two sub-cases for $\Phi_1(b, a) = (b + p_1) + \max(b + p_1, a) + q_1$:

If $a \geq b + p_1$: $\Phi_1 = b + p_1 + a + q_1 = a + b + p_1 + q_1 \leq a + b + 2p_1 + q_1$. ✓

If $a < b + p_1$: $\Phi_1 = 2(b + p_1) + q_1 = 2b + 2p_1 + q_1 \leq a + b + 2p_1 + q_1$ since $b < a$. ✓

Case 2c ($b + q_1 > a$, B-first): $\text{sum} = (b + q_1) + (b + p_1 + q_1) = 2b + p_1 + 2q_1$.

If $a \geq b + p_1$: $\Phi_1 = a + b + p_1 + q_1 \leq 2b + p_1 + 2q_1$ since $a \leq b + q_1$ (overlap condition) gives $a - b \leq q_1$. ✓

If $a < b + p_1$: $\Phi_1 = 2b + 2p_1 + q_1 \leq 2b + p_1 + 2q_1$ since $p_1 \leq q_1$. ✓

Inductive step: $L - 1 \rightarrow L$

Assume both statements for $(L - 1)$ -stage chains.

Proposition (P) for L stages. Arrivals (α, β) with $\alpha \leq \beta$. Schedule σ uses priority π_1 at stage 1.

$\Phi_L(\alpha, \beta) = \Phi_{L-1}(\alpha + p_1, \max(\alpha + p_1, \beta) + q_1)$.

Case A: $\pi_1 = A\text{-first}$. Stage-1 outputs: $C_1(A) = \alpha + p_1$, $C_1(B) = \max(\alpha + p_1, \beta) + q_1$, with $C_1(A) \leq C_1(B)$. By Proposition (P) (inductive, $L - 1$ stages): $C_L^\sigma \geq \Phi_{L-1}(C_1(A), C_1(B)) = \Phi_L(\alpha, \beta)$. ✓

Case B: $\pi_1 = B\text{-first}$.

No overlap ($\alpha + p_1 \leq \beta$): same output as A-first (Case 1a). Reduces to Case A. ✓

Overlap ($\alpha + p_1 > \beta$): stage-1 outputs $C_1(A) = \alpha + p_1 + q_1$, $C_1(B) = \beta + q_1$, with $C_1(A) > C_1(B)$ (B first at stage 2).

By Lemma (Q) (inductive, $L - 1$ stages) with arrivals $(\alpha + p_1 + q_1, \beta + q_1)$:

$$C_L^\sigma(A) + C_L^\sigma(B) \geq \Phi_{L-1}(\beta + q_1, \alpha + p_1 + q_1).$$

Since $\alpha + p_1 > \beta$ (overlap), $\Phi_L(\alpha, \beta) = \Phi_{L-1}(\alpha + p_1, \alpha + p_1 + q_1)$.

The second arguments are equal ($= \alpha + p_1 + q_1$). Since $\alpha + p_1 \leq \beta + q_1$ (because $\alpha \leq \beta$ and $p_1 \leq q_1$), monotonicity of Φ_{L-1} in the first argument gives $\Phi_{L-1}(\alpha + p_1, \alpha + p_1 + q_1) \leq \Phi_{L-1}(\beta + q_1, \alpha + p_1 + q_1)$.

Therefore $\Phi_L(\alpha, \beta) \leq C_L^\sigma$. ✓

Lemma (Q) for L stages. Arrivals (a, b) with $a > b$. Schedule σ uses priority π_1 at stage 1. We must show $C_L^\sigma \geq \Phi_L(b, a)$.

Write $\Phi_L(b, a) = \Phi_{L-1}(b + p_1, \max(b + p_1, a) + q_1)$.

Define $\mu = \max(b + p_1, a)$ and note $\Phi_L(b, a) = \Phi_{L-1}(b + p_1, \mu + q_1)$.

Case 2a: no overlap ($b + q_1 \leq a$).

Both priorities give $C_1(A) = a + p_1$, $C_1(B) = b + q_1$, with $C_1(A) > C_1(B)$ (B first).

By Lemma (Q) (inductive, $L - 1$ stages): $C_L^\sigma \geq \Phi_{L-1}(b + q_1, a + p_1)$.

Since $b + q_1 \leq a$ implies $b + p_1 \leq a$, we have $\mu = a$ and $\Phi_L(b, a) = \Phi_{L-1}(b + p_1, a + q_1)$.

Both pairs $(b + q_1, a + p_1)$ and $(b + p_1, a + q_1)$ have the same sum $T = a + b + p_1 + q_1$, with first arguments $b + q_1 \geq b + p_1$ (since $q_1 \geq p_1$) and $b + q_1 \leq a + p_1$ (since $b + q_1 \leq a \leq a + p_1$). By Lemma A.5:

$$\Phi_{L-1}(b + q_1, a + p_1) \geq \Phi_{L-1}(b + p_1, a + q_1) = \Phi_L(b, a). \quad \checkmark$$

Case 2b: overlap, A-first ($b + q_1 > a$, $\pi_1 = A$).

Outputs: $C_1(A) = a + p_1$, $C_1(B) = b + p_1 + q_1$. Since $b + q_1 > a$: $C_1(B) = b + p_1 + q_1 > a + p_1 = C_1(A)$, so A arrives first at stage 2.

By Proposition (P) (inductive, $L - 1$ stages): $C_L^\sigma \geq \Phi_{L-1}(a + p_1, b + p_1 + q_1)$.

We show $\Phi_{L-1}(a + p_1, b + p_1 + q_1) \geq \Phi_L(b, a) = \Phi_{L-1}(b + p_1, \mu + q_1)$.

Sub-case $a \geq b + p_1$ ($\mu = a$): $\Phi_L(b, a) = \Phi_{L-1}(b + p_1, a + q_1)$.

The pairs $(a + p_1, b + p_1 + q_1)$ and $(b + p_1, a + p_1 + q_1)$ have the same sum $T' = a + b + 2p_1 + q_1$, with $a + p_1 \geq b + p_1$. Both first arguments are $\leq$ the respective second arguments (the first pair: $a + p_1 \leq b + p_1 + q_1$ since $a < b + q_1$; the second: $b + p_1 \leq a + p_1 + q_1$ trivially). By Lemma A.5:

$$\Phi_{L-1}(a + p_1, b + p_1 + q_1) \geq \Phi_{L-1}(b + p_1, a + p_1 + q_1).$$

By monotonicity of Φ_{L-1} in the second argument ($a + p_1 + q_1 \geq a + q_1$):

$$\Phi_{L-1}(b + p_1, a + p_1 + q_1) \geq \Phi_{L-1}(b + p_1, a + q_1) = \Phi_L(b, a). \quad \checkmark$$

Sub-case $a < b + p_1$ ($\mu = b + p_1$): $\Phi_L(b, a) = \Phi_{L-1}(b + p_1, b + p_1 + q_1)$.

Since $a > b$: $a + p_1 > b + p_1$. By monotonicity in the first argument:

$$\Phi_{L-1}(a + p_1, b + p_1 + q_1) \geq \Phi_{L-1}(b + p_1, b + p_1 + q_1) = \Phi_L(b, a). \quad \checkmark$$

Case 2c: overlap, B-first ($b + q_1 > a$, $\pi_1 = B$).

Outputs: $C_1(B) = b + q_1$, $C_1(A) = b + p_1 + q_1$. Since $p_1 > 0$: $C_1(A) > C_1(B)$, so B first at stage 2.

By Lemma (Q) (inductive, $L - 1$ stages): $C_L^\sigma \geq \Phi_{L-1}(b + q_1, b + p_1 + q_1)$.

We show $\Phi_{L-1}(b + q_1, b + p_1 + q_1) \geq \Phi_L(b, a) = \Phi_{L-1}(b + p_1, \mu + q_1)$.

Sub-case $a \geq b + p_1$ ($\mu = a$): $\Phi_L(b, a) = \Phi_{L-1}(b + p_1, a + q_1)$.

We split into two sub-sub-cases according to whether $b + q_1 \leq a + p_1$ or $b + q_1 > a + p_1$.

If $b + q_1 \leq a + p_1$: both pairs $(b + q_1, b + p_1 + q_1)$ and $(b + p_1, a + q_1)$ have sum $T = a + b + p_1 + q_1$ (for the second pair) and $T' = 2b + p_1 + 2q_1 > T$ (for the first, since $b + q_1 > a$). By monotonicity in the second argument: $\Phi_{L-1}(b + q_1, b + p_1 + q_1) \geq \Phi_{L-1}(b + q_1, a + p_1)$ (since $b + p_1 + q_1 \geq a + p_1$ as $b + q_1 > a$). Both pairs $(b + q_1, a + p_1)$ and $(b + p_1, a + q_1)$ now have sum T , with $b + q_1 \leq a + p_1$ (assumption) and $b + q_1 \geq b + p_1$ (since $q_1 \geq p_1$). By Lemma A.5: $\Phi_{L-1}(b + q_1, a + p_1) \geq \Phi_{L-1}(b + p_1, a + q_1) = \Phi_L(b, a)$.
✓

If $b + q_1 > a + p_1$: both pairs $(a + p_1, b + q_1)$ and $(b + p_1, a + q_1)$ have the same sum $T = a + b + p_1 + q_1$ and satisfy first $\leq$ second. Since $a + p_1 \geq b + p_1$ (as $a > b$), Lemma A.5 gives $\Phi_{L-1}(a + p_1, b + q_1) \geq \Phi_{L-1}(b + p_1, a + q_1) = \Phi_L(b, a)$.

It remains to show $\Phi_{L-1}(b + q_1, b + p_1 + q_1) \geq \Phi_{L-1}(a + p_1, b + q_1)$. Since $b + q_1 \geq a + p_1$ (assumption) and $b + p_1 + q_1 \geq b + q_1$ (as $p_1 \geq 0$), both arguments of the left side are at least as large as the corresponding arguments on the right. By Lemma A.4: the inequality holds.

Combining: $\Phi_{L-1}(b + q_1, b + p_1 + q_1) \geq \Phi_{L-1}(a + p_1, b + q_1) \geq \Phi_{L-1}(b + p_1, a + q_1) = \Phi_L(b, a)$.
✓

Sub-case $a < b + p_1$ ($\mu = b + p_1$): $\Phi_L(b, a) = \Phi_{L-1}(b + p_1, b + p_1 + q_1)$.

Since $q_1 \geq p_1$: $b + q_1 \geq b + p_1$. By monotonicity in the first argument: $\Phi_{L-1}(b + q_1, b + p_1 + q_1) \geq \Phi_{L-1}(b + p_1, b + p_1 + q_1) = \Phi_L(b, a)$. ✓

This completes the inductive step for both Proposition (P) and Lemma (Q), proving Theorem 23. □

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